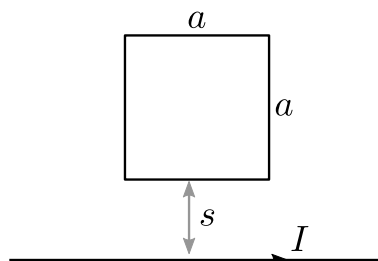


Homework assignment for 02YELMA

Chapter: Electrodynamics



1. A square wire loop lies a distance s from a very long straight wire that holds a current I as shown above. Find the flux of magnetic field Φ through the loop. If the wire is pulled directly away from the wire at speed v what emf is generated and in what direction does the current flow?
2. A long solenoid of radius a is driven by an alternating current so that the field inside is sinusoidal: $\vec{B}(t) = B_0 \cos(\omega t) \hat{z}$. A circular loop of wire of radius $a/2$ and resistance R is placed inside the solenoid and coaxial with it. Find the current induced in the loop as a function of time.
3. Find the self-inductance per unit length of a long solenoid, of radius R , carrying n turns per unit length.
4. Find the self-inductance per unit length of a long solenoid of radius R with n turns per unit length.
5. A *sufficiently long* solenoid with radius a and n turns per unit length carries a time-dependent current $I(t)$ in the $\hat{\phi}$ direction. Find the electric field (direction and magnitude) at a distance s from the axis of the solenoid (both inside and outside the solenoid) in the *quasi-static approximation*.
 - *Sufficiently long*: There is a uniform magnetic field inside the solenoid and negligible field outside.
 - *Quasi-static approximation*: The change in the magnetic field is slow enough that the rules of magnetostatics apply to determine the magnetic field at a given time.
6. A parallel-plate capacitor of plate radius R and plate separation d is being driven by a sinusoidal voltage $V(t) = V_0 \cos(\omega t)$ across its plates. The space between the plates is vacuum.
 - (a) Compute the displacement current $\vec{J}_d(t)$ flowing between the plates.
 - (b) Compute the magnitude of magnetic field $B(r, t)$ at a distance from the capacitor axis.